Erdős problem #1056 as a moment problem for factorials modulo p

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Abstract

Erdős problem #1056 asks, for each $k \geq 2$, for a prime p and k consecutive integer intervals each of product $\equiv 1 \pmod{p}$ — equivalently, for $k+1$ residues in $[1, p-1]$ sharing a common factorial value modulo p. All known unconditional *infinitude* results operate at the pinned common values ± 1, where Wilson-type identities apply, and [10] proves that toolkit terminates at $k = 3$. This note records an elementary but consequential translation: for every k there is an explicit $r = r(k)$ such that a lower bound on the r-th moment of the multiplicity function of $n! \bmod p$, held on an infinite set of primes, implies #1056 at k with infinitely many prime witnesses — at a *free* common value, entirely outside the scope of the pinned-value obstruction. Under the standard Poisson heuristic for factorials modulo p the hypothesis holds for every k with an explicit margin. The first case not already known unconditionally is $r(4) = 12$: a lower bound $V_{12}(p) > 4^{11}(p-1)$ on infinitely many primes implies the infinitude of pentad primes. We state the calibration table, the supporting census (97.7% of the primes $7 \leq p < 20000$ carry $k \geq 4$ at some value; only 15.2% at value 1), and the governing conjecture, and we pose the minimal open analytic targets.

1 The bridge

Fix an odd prime p. For $c \in \mathbb{F}_p^\times$ let

$$\mu_p(c) \;=\; \#\{\, n \in [1, p-1] : n! \equiv c \pmod{p} \,\}$$

With machine-assisted proof search and verification. All computations reported here, and substantial portions of the proof search and adversarial checking, were carried out by AI agents (Claude, Anthropic) under the author's direction. Companion to [10], whose Remark 6.1-style provenance disclosure applies here as well.

be the multiplicity function of the factorial map, and for $r \geq 1$ let

$$V_r(p) \;=\; \sum_{c \in \mathbb{F}_p^\times} \mu_p(c)^r \;=\; \#\{\,(n_1, \ldots, n_r) \in [1, p-1]^r : n_1! \equiv \cdots \equiv n_r! \pmod p\,\}$$

be its r-th moment. Since $n! \not\equiv 0$ for $n \leq p-1$, we have $\sum_c \mu_p(c) = p-1$ exactly. Throughout, intervals are $I_i = (q_{i-1}, q_i]$ with $1 \leq q_0 < q_1 < \cdots < q_k \leq p-1$: this is the equal-factorials form of Noll and Simmons (see [7] and Guy [9], A15), the convention recorded by OEIS A060427, and it excludes the degenerate leading block that $0! = 1$ would otherwise supply. Write $k(p)$ for the largest such k; then $k(p)+1 = \max_c \mu_p(c)$, as is immediate: the interval $(q_i, q_{i+1}]$ has unit product if and only if $q_i! \equiv q_{i+1}! \pmod p$.

Lemma 1.1 (Moment bridge)**.** *Let $k \geq 1$ and $r \geq 2$. If $V_r(p) > k^{r-1}(p-1)$, then $k(p) \geq k$. Consequently, if $V_r(p) > k^{r-1}(p-1)$ for infinitely many primes p, then Erdős #1056 holds at k with infinitely many prime witnesses.*

Proof. Suppose $\max_c \mu_p(c) \leq k$. Then $\mu_p(c)^r \leq k^{r-1}\mu_p(c)$ for every c, so $V_r(p) \leq k^{r-1} \sum_c \mu_p(c) = k^{r-1}(p-1)$. The contrapositive gives some c with $\mu_p(c) \geq k+1$: that is $k+1$ distinct elements of $[1, p-1]$ with a common factorial value, hence k consecutive intervals of unit product. The bound is uniform in p, so infinitely many qualifying p give infinitely many witnesses. □

The lemma is pigeonhole and makes no claim to depth. Its content is the *translation*: it converts #1056 from "produce a rare arithmetic configuration" into "beat an explicit constant in an averaged count", restating the problem in the language of moments and multiplicative energy, where analytic technology operates. Two features deserve emphasis.

First, the common value c produced by the lemma is *free*. The limitative theorem of [10] shows that the classical identity toolkit cannot pass $k = 3$ *at the pinned value* 1; the configurations supplied by Lemma 1.1 carry no value constraint and are untouched by that obstruction. The census below shows free values are where almost all large-multiplicity configurations actually live.

Second, the bound in the lemma is tight (all mass at multiplicity exactly k attains it), so the exponent r must be chosen large enough that the *expected* moment clears the threshold. That calibration is the next section.

2 Calibration under the Poisson model

The standard heuristic — conjectured explicitly by Cobeli and Zaharescu [5], generalizing Stauduhar, with the random-model analogue proved by Cobeli, Vâjâitu, and Zaharescu [4], and consistent with all computations we are aware of, including the census below — is that for large p the multiset $\{n! \bmod p : 1 \leq n \leq p-1\}$ behaves like $p-1$ values drawn uniformly at random from $\mathbb{F}_p^\times$, so that the multiplicities $\mu_p(c)$ behave like independent Poisson random variables of mean 1. The r-th moment of a Poisson(1) variable is the r-th Bell number B_r (Dobiński's formula), so the model predicts

$$V_r(p) \;\sim\; B_r\,(p-1).$$

Define

$$r(k) \;=\; \min\{\, r : B_r > k^{r-1} \,\},$$

which exists for every k since B_r grows super-exponentially in r. Under the model, the hypothesis of Lemma 1.1 at $r = r(k)$ holds for all large p with margin $B_{r(k)}/k^{r(k)-1}$:

k	$r(k)$	margin $B_{r(k)}/k^{r(k)-1}$
2	3	1.250
3	7	1.203
4	12	1.005
5	19	1.529
6	26	1.746
7	33	1.476
8	41	1.769
9	49	1.686
10	57	1.295

(Continuing: $r(11), \ldots, r(15) = 66, 75, 84, 93, 103$.) All values were computed exactly from the Bell triangle. Note the conspicuously thin margin at $k = 4$: the pentad rung $r = 12$ clears its threshold by only 0.5%, with $B_{12} = 4{,}213{,}597$ against $4^{11} = 4{,}194{,}304$. Any proof attempt at $k = 4$ via $r = 12$ has essentially no room; the first comfortable rung at $k = 4$ is $r = 13$ ($B_{13}/4^{12} = 1.648$), at the price of a higher moment.

3 What is known unconditionally

The ladder below Lemma 1.1 is remarkably short, which calibrates how early the unconditional theory of $n! \bmod p$ still is.

- $k(p) \geq 1$ **for all** $p \geq 5$ is classical: Wilson's theorem gives $(p-2)! \equiv 1 \equiv 1!$, the universal pair.

- **Non-injectivity on** $[2, p-1]$ — the collision question with the universal pair excluded, i.e. the nonexistence of "socialist primes" $p > 5$. Rokowska and Schinzel [13] proved restrictive congruence conditions in 1960; nonexistence was verified computationally to 10^9 by Trudgian [14] (who coined the term) and to 10^{11} by Andrejić and Tatarević [2]. A proof for all $p > 5$ has recently been announced by Abramov [1]; the preprint is elementary and, to our knowledge, not yet independently verified.

- $k(p) \geq 2$ **for all large** p **is open**, to our knowledge. Empirically the exceptional set is tiny and appears complete very early: the only odd primes $p < 20000$ with $k(p) \leq 1$ are

$$p \in \{3,\ 5,\ 7,\ 13,\ 19,\ 37\},$$

  so $k(p) \geq 2$ for every prime $41 \leq p < 20000$. We know of no proof of $k(p) \geq 2$ valid for all large p.

- **Existence for** $2 \leq k \leq 14$: OEIS A060427 [12]; the record witness is $a(14) = 10428007$ (each record witnesses all smaller k). An exhaustive scan by the author shows $a(15) > 3.2 \times 10^7$.

- **Infinitude for** $k \leq 3$: the tetrad theorem of Agustín-Aquino and Hernández Santiago [6], at the pinned value 1; [10] adds the pentad criterion (along the family $p \equiv 3 \bmod 4$, $p \mid q! - 1$, $q \equiv 1 \bmod 6$, the central residue $m = (p-1)/2$ joins the tetrad $\{1, q, p-1-q, p-2\}$ if and only if $h(-p) \equiv 3 \bmod 4$) and proves the pinned-value toolkit terminates there.

- **Distributional results**: Klurman and Munsch [11] study the image of $n! \bmod p$ in short intervals, and the number of residue classes it misses on average over p, in the direction of the random model. Missing-value results already give second-moment excess: by the identity $\sum_c (\mu_p(c) - 1)^+ = \#\{\text{missed classes}\}$, the unconditional lower bound of Banks, Luca, Shparlinski, and Stichtenoth [3] on missed classes yields $V_2(p) \geq (p-1) + \Omega(\log\log p / \log\log\log p)$ on an infinite set of primes, and [11] improves the missed-class count on average (and under GRH). In the opposite direction, character-sum technology

(Garaev–Luca–Shparlinski [8] and its successors) gives *upper*-bound-flavored control on congruences with $n!$. All known moment excesses are $o(p)$: no bound $V_r(p) \geq c_r(p-1)$ with a constant $c_r > 1$ appears to be known for any $r \geq 2$.

4 The census, and where the truth lives

Direct computation over the 2259 primes $7 \leq p < 20000$ ([10], Remark on free common values, and the verification code in the repository):

- 2207 primes (97.7%) satisfy $k(p) \geq 4$ at *some* common value;
- only 344 (15.2%) do so at the value 1, the only regime where any current technique operates;
- the maximum multiplicity grows steadily: e.g. $p = 977$ carries six equal factorials at the value 725 ($k = 5$, $p \equiv 1 \bmod 4$, no class number involved).

The wall at $k = 3$ is thus a wall of *provability*, not of scarcity. The Poisson model predicts far more: the maximum of $\sim p$ independent Poisson(1) multiplicities is the classical maximum-load statistic,

$$\max_c \mu_p(c) \;\sim\; \frac{\log p}{\log\log p},$$

which motivates:

Conjecture 4.1. *$k(p) \to \infty$ as $p \to \infty$. Quantitatively, $k(p) = (1 + o(1))\log p/\log\log p$ for almost all p.*

Conjecture 4.1 implies the full Erdős #1056 (every k is witnessed, indeed by all sufficiently large primes outside a density-zero set). Erdős already suggested that witnesses exist for every k [9]; the conjecture upgrades this to almost-all p with the Poisson maximum-load rate, completing to all multiplicities the random model for factorials discussed in [11] and [5].

5 The open targets

We close with the precise statements the translation makes available, in increasing order of consequence.

Question 5.1. Is $k(p) \geq 2$ for all sufficiently large p? Equivalently: does every large prime admit three equal factorial residues? It holds for infinitely many p [6], and unconditionally for every $p \equiv 3 \bmod 4$ with $h(-p) \equiv 3 \bmod 4$, where Mordell's identity puts $\{1, (p-1)/2, p-2\}$ at value 1 [10]; the open content is the all-but-finitely-many form. (The moment version: $V_3(p) > 4(p-1)$ would suffice for the primes on which it holds; the model predicts $V_3(p) \sim 5(p-1)$.)

Question 5.2. Is $V_{12}(p) > 4^{11}(p-1)$ for infinitely many p? By Lemma 1.1 this implies infinitely many pentad primes — at a free common value, bypassing both the class number hypothesis and the simultaneous-divisibility hypotheses of [10]. The same conclusion follows from $V_{13}(p) > 4^{12}(p-1)$, where the model's margin is 1.648 rather than 1.005.

Question 5.3. Is there *any* $r \geq 2$ and any constant $c_r > 1$ with $V_r(p) \geq c_r\,(p-1)$ for infinitely many p? The known excesses are additive and small: (2^r-2) from the universal pair ($p \geq 5$), and $\Omega(\log\log p/\log\log\log p)$ at $r = 2$ from the missed-class bound of [3] via $\sum_c (\mu_p(c) - 1)^+ = \#\{\text{missed classes}\}$. All of these are $o(p)$; no constant-factor bound is known for any r, while the model predicts the constant B_r. Any such bound would be the first, and the method would matter more than the constant.

Remark 5.4 (Connection to the image size). Write $N(p) = \#\{n! \bmod p : 1 \leq n \leq p-1\}$ for the image size — the central quantity of [11], long conjectured (Erdős–Graham; see Guy [9] and the discussion in [11]) to satisfy $N(p) \sim (1 - e^{-1})\,p$. By Cauchy–Schwarz, $V_2(p) \geq (p-1)^2/N(p)$, so any unconditional upper bound $N(p) \leq (1-\delta)(p-1)$ would resolve Question 5.3 at $r = 2$ with $c_2 = 1/(1-\delta)$ — though not conversely: a single value of multiplicity $\sim \sqrt{cp}$ makes V_2 large while leaving $N(p) = (1 - o(1))(p-1)$. The missing lower bounds on moments are thus implied by, and at most as hard as, the missing upper bounds on the image — a second reason the moment formulation sits squarely in the territory of [11].

We make no claim that these are easy: lower bounds on the multiplicative energy of $\{n!\}$ modulo p are exactly what current exponential-sum technology does not provide, and the limitative theorem of [10] explains why the algebraic-identity route cannot substitute for them beyond $k = 3$. The point of this note is narrower: Erdős #1056 is, after Lemma 1.1, a well-posed moment problem with an explicit finite target for each k, a calibrated random model that clears every target with stated margin, and a first collision statement beyond Wilson recently claimed by Abramov [1]. It is now a problem stated entirely in the vocabulary of [11] and its antecedents.

Remark 5.5 (Verification status). The Bell numbers, the $r(k)$ table, the census figures, the exceptional set $\{3, 5, 7, 13, 19, 37\}$, and the $a(15)$ bound were each computed at least twice in independently written code. Lemma 1.1 is elementary; no computer verification is claimed or needed for it.

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